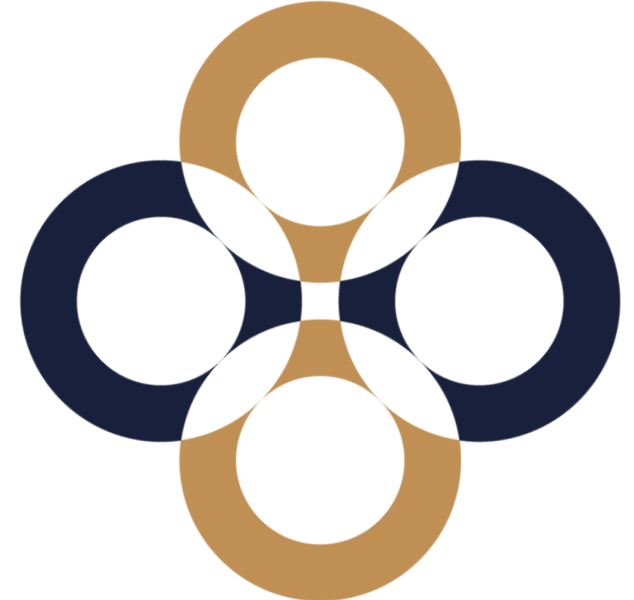


Risk and Return tradeoff, CAPM

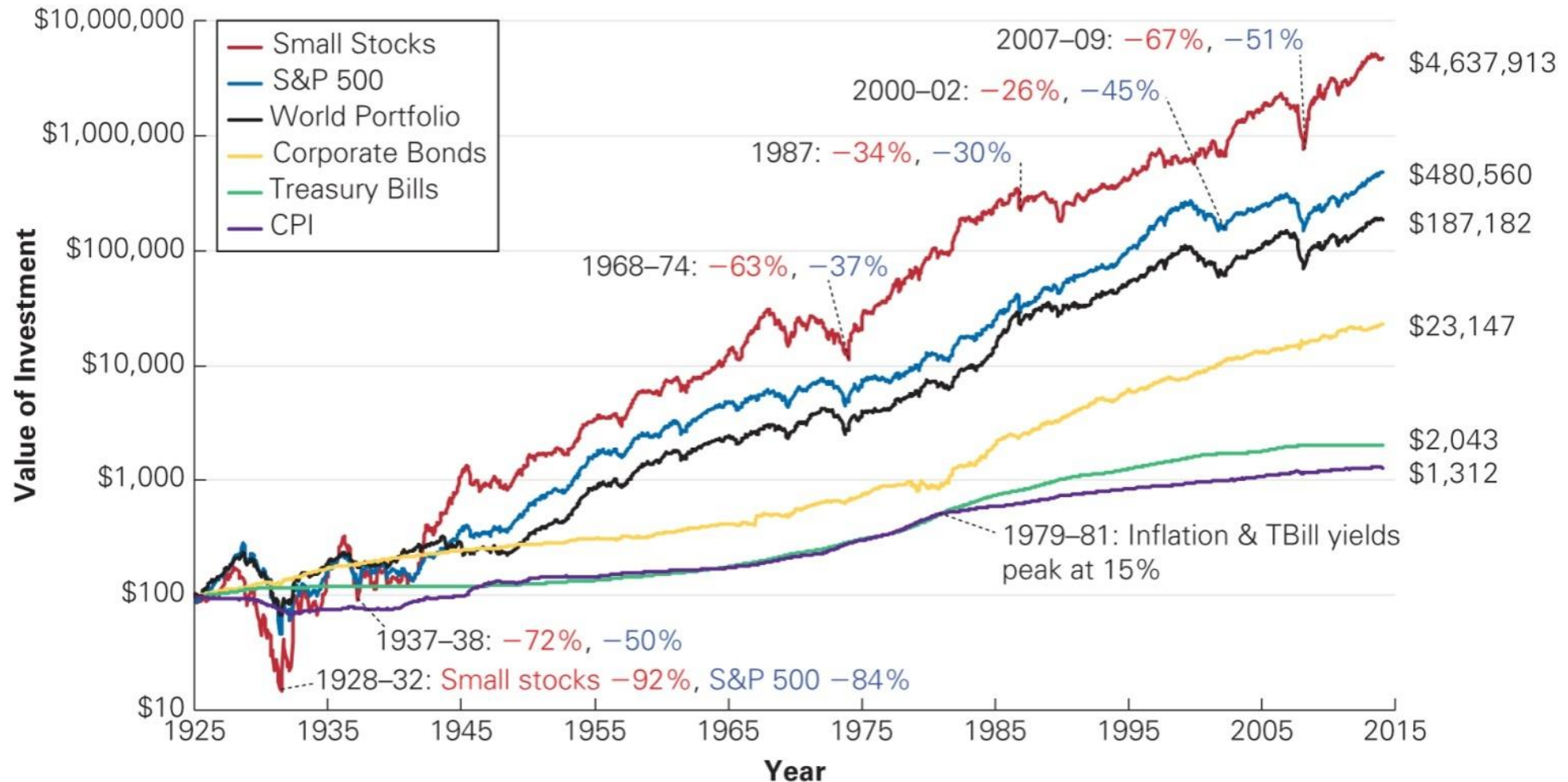
International Corporate Finance – 3

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Corvinus University of Budapest, Institute of Finance

Based on:
Berk – DeMarzo (2017): Corporate Finance. Fourth edition, Pearson – chapters 10-12



Value of \$100 Invested at the End of 1925



Common Measures of Risk and Return

- Probability Distributions
 - Each possible return has some likelihood of occurring.
 - This information is summarized with a probability distribution:
 - a probability, P_R , that each possible return, R , will occur.
 - Assume BFI stock currently trades for \$100 per share. In one year, there is a 25% chance the share price will be \$140, a 50% chance it will be \$110, and a 25% chance it will be \$80.

Common Measures of Risk and Return

- Expected (Mean) Return

$$\text{Expected Return} = E[R] = \sum_R P_R \times R$$

- Variance

The expected squared deviation from the mean

$$\text{Var}(R) = E\left[\left(R - E[R]\right)^2\right] = \sum_R P_R \times \left(R - E[R]\right)^2$$

- Standard Deviation (volatility)

$$SD(R) = \sqrt{\text{Var}(R)}$$

Expected return, variance, covariance, correlation

Expected return (average):

$$\bar{r}_x = p_1 \cdot r_1 + p_2 \cdot r_2 + p_3 \cdot r_3 + \dots + p_T \cdot r_T = \sum_i p_i \cdot r_i = \frac{\sum_i r_i}{T} \rightarrow -\infty < = \bar{r}_x < = \infty$$

Variance:

$$Var(r_x) = \frac{\sum_i (r_i - \bar{r}_x)^2}{T} \rightarrow \sqrt{Var(r_x)} = \sigma_x \rightarrow \mathbf{Var} \geq 0$$

Covariance:

$$Cov(r_x, r_y) = \frac{\sum_i (r_{xi} - \bar{r}_x)(r_{yi} - \bar{r}_y)}{T} \rightarrow -\infty < = \mathbf{Cov} < = \infty$$

Correlation:

$$Corr(r_x, r_y) = \frac{Cov(r_x, r_y)}{\sigma_x \sigma_y} \rightarrow \mathbf{-1 < = Corr < = 1}$$

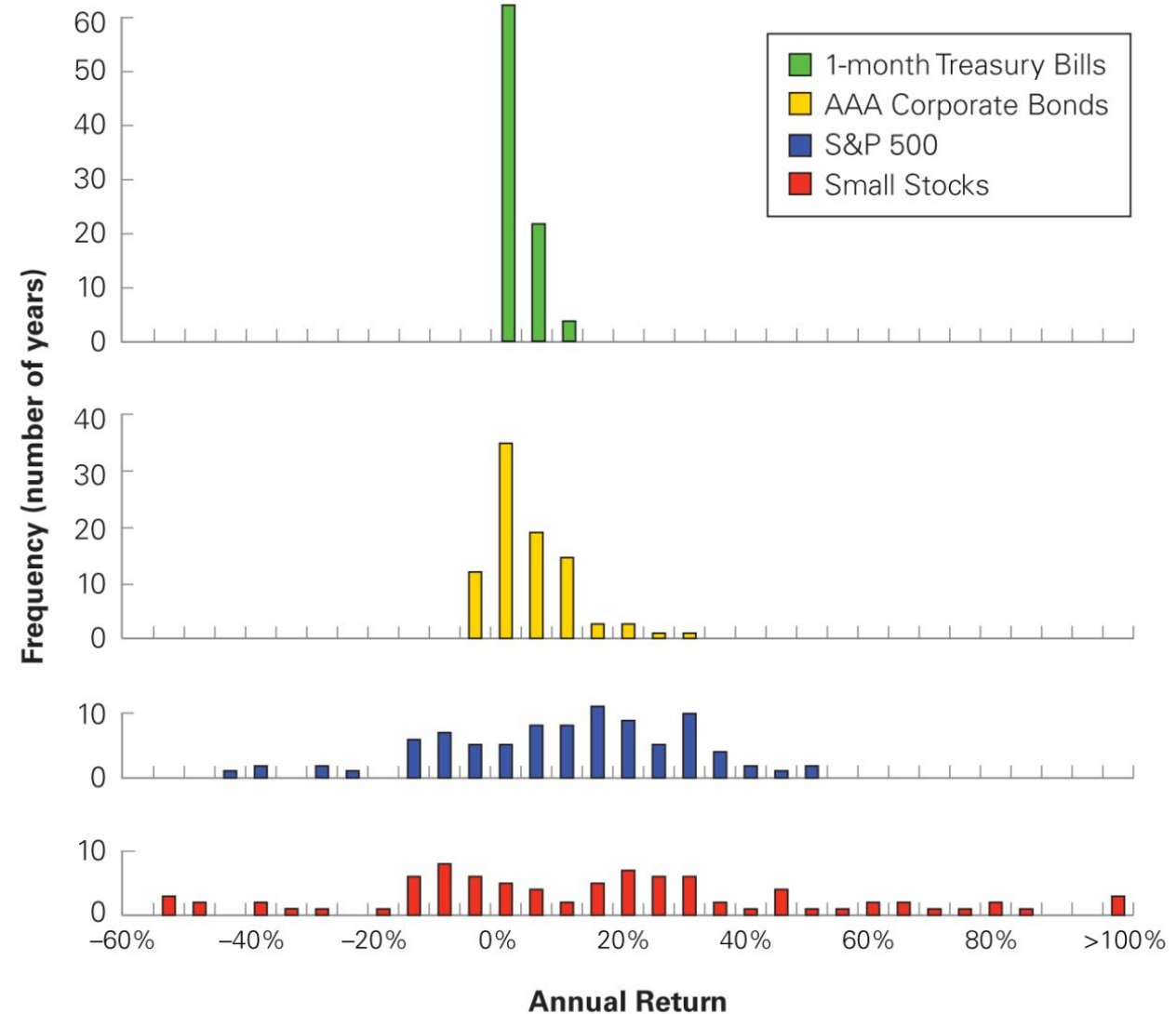
→ or (T-1) if they are estimated from a sample (above 100 observations I assume it is the whole population)



The Empirical Distribution of Annual Returns 1926–2014

9/20/2024

- Empirical Distribution
 - When the probability distribution is plotted using historical data



Using Past Returns to Predict the Future

- Average annual return

$$\bar{R} = \frac{1}{T} (R_1 + R_2 + \dots + R_T) = \frac{1}{T} \sum_{t=1}^T R_t$$

- Variance Estimate Using Realized Returns

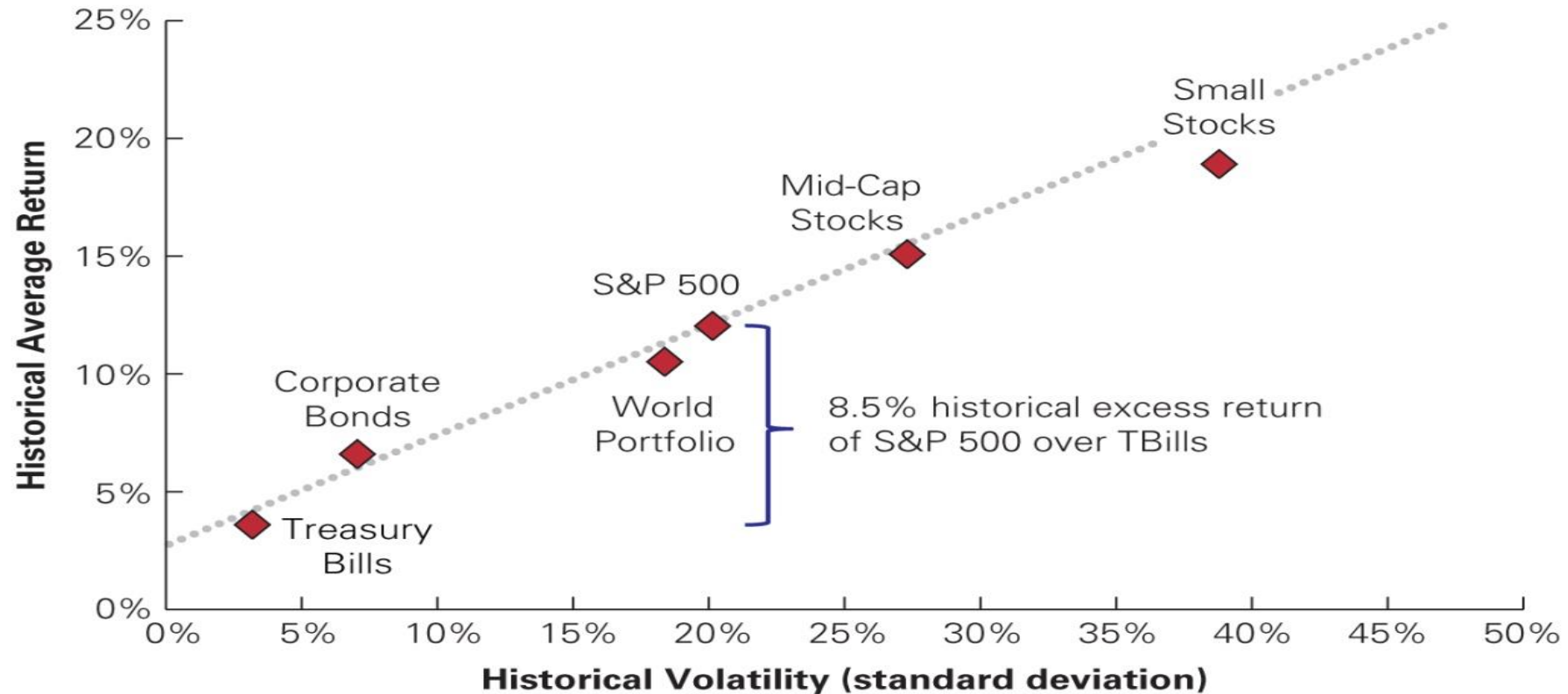
$$Var(R) = \frac{1}{T - 1} \sum_{t=1}^T (R_t - \bar{R})^2$$

- Standard Error

- A statistical measure of the degree of estimation error

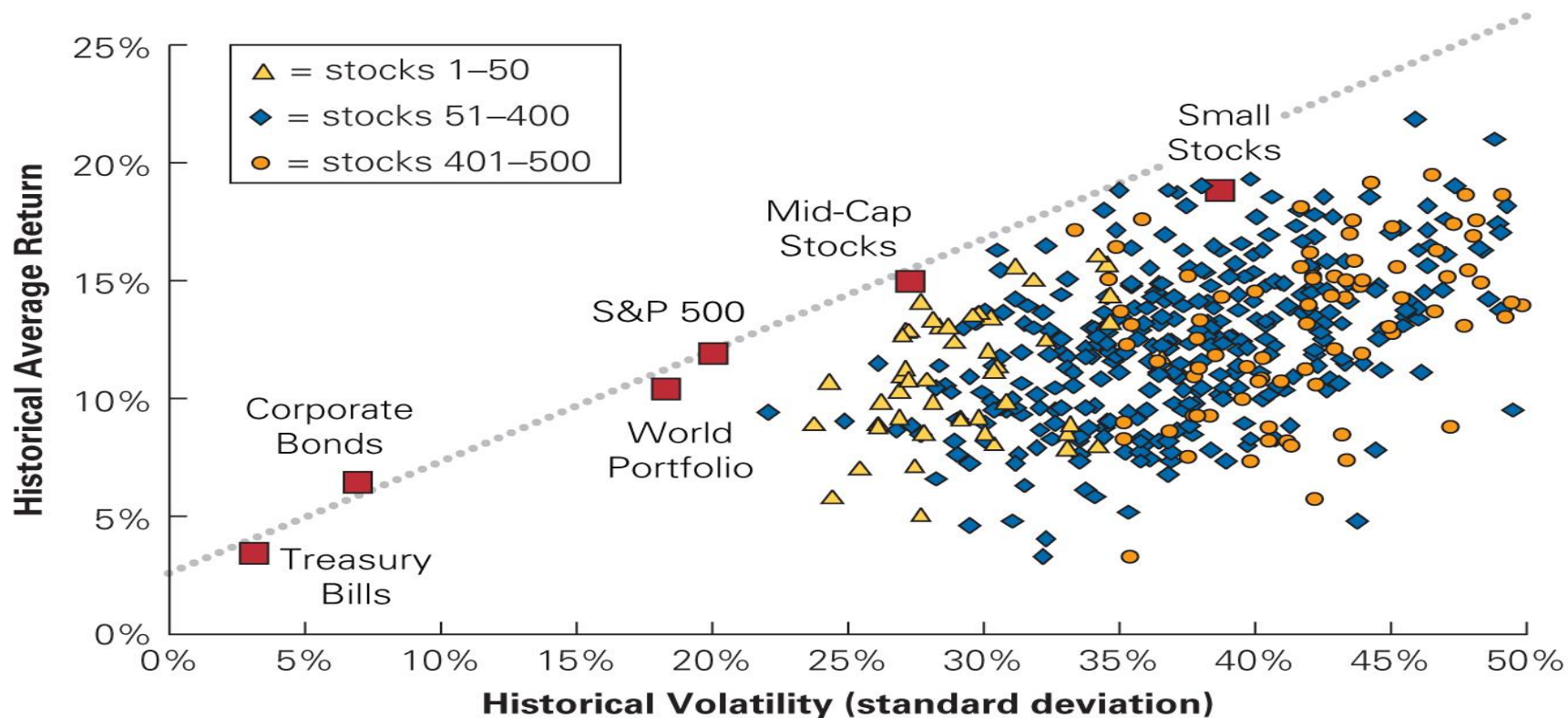
$$SD(\text{Average of Independent, Identical Risks}) = \frac{SD(\text{Individual Risk})}{\sqrt{\text{Number of Observations}}}$$

The Historical Trade-Off Between Risk and Return in Large Portfolios



Source: CRSP, Morgan Stanley Capital International

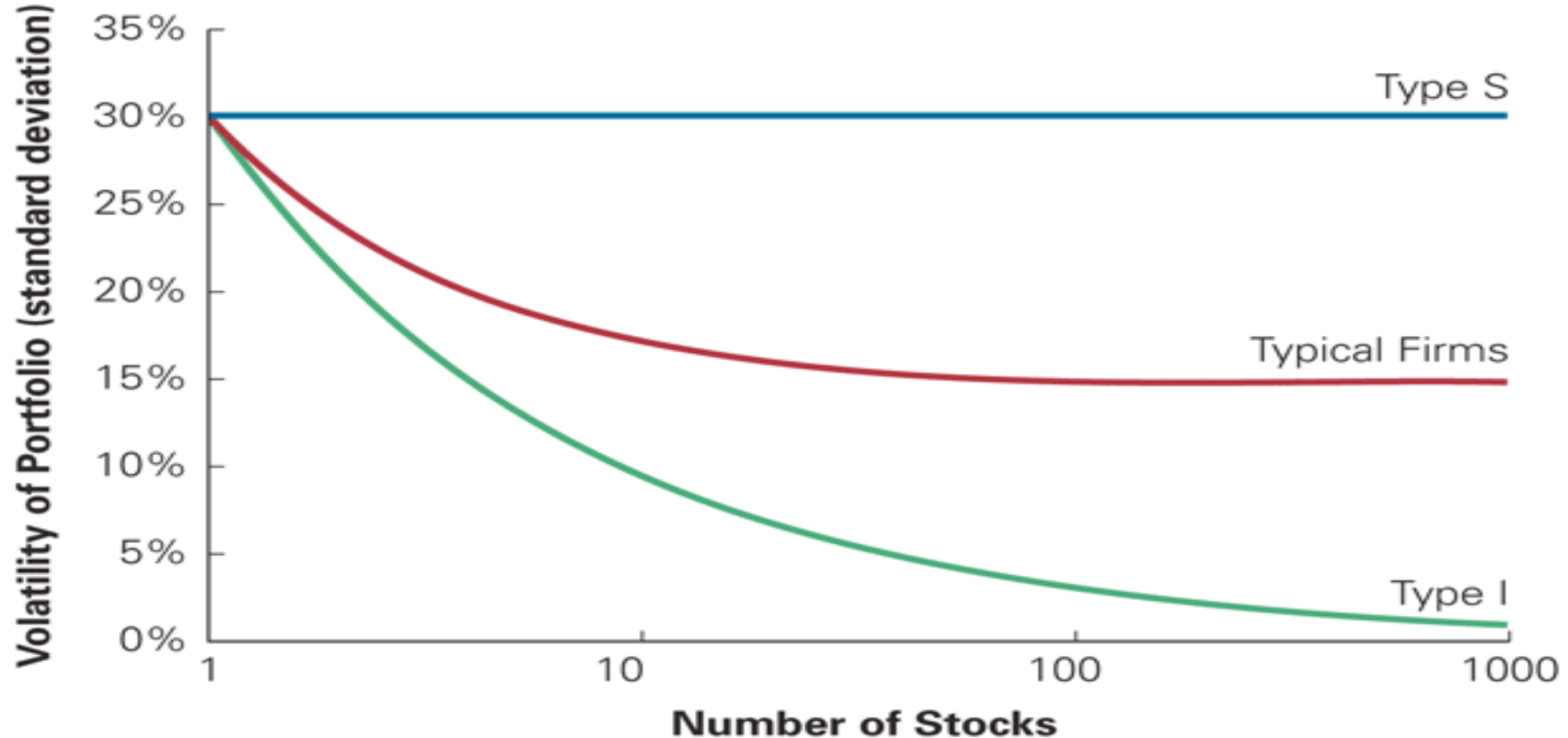
Historical Volatility and Return for 500 Individual Stocks, Ranked Annually by Size



Source: CRSP

- Common (systematic, undiversifiable , market) Risk
 - Risk that is perfectly correlated
 - Risk that affects all securities
 - Due to Market-Wide News
- Independent (firm-specific, idiosyncratic, unique, unsystematic, diversifiable) Risk
 - Risk that is uncorrelated
 - Risk that affects a particular security
 - Due to Firm Specific News
- Diversification
 - The averaging out of independent risks in a large portfolio

Volatility of Portfolios of Type S and I Stocks



Stock's volatility (total risk)=Diversifiable risk+Systematic risk

- **The risk premium for diversifiable risk is zero**
- **The risk premium of a security is determined by its systematic risk**
- **Volatility is not especially useful in determining the risk premium**
- No clear relationship between volatility and average returns for individual securities.
- **Security's sensitivity to systematic risk:**
 - average change in the return for each 1% change in the return of a portfolio that fluctuates solely due to systematic risk

Measuring Systematic Risk

- Efficient Portfolio
 - A portfolio that contains only systematic risk. There is no way to reduce the volatility of the portfolio without lowering its expected return.
- Market Portfolio
 - An efficient portfolio that contains all shares and securities in the market
 - The S&P 500 is often used as a proxy for the market portfolio.
- Sensitivity to Systematic Risk: Beta (β)
 - **The expected percent change in the excess return of a security for a 1% change in the excess return of the market portfolio.**
- Stocks in cyclical industries are likely to be more sensitive to systematic risk (higher beta).

Beta and the Cost of Capital

- Market risk premium
 - The market risk premium is the reward investors expect to earn for holding a portfolio with a beta of 1

$$\text{Market Risk Premium} = E[R_{Mkt}] - r_f$$

- Security's Cost of Capital of an investment from Its Beta

$$\begin{aligned} E[R] &= \text{Risk-Free Interest Rate} + \text{Risk Premium} \\ &= r_f + \beta \times (E[R_{Mkt}] - r_f) \end{aligned}$$

- **Capital Asset Pricing Model (CAPM).**

The Expected Return of a Portfolio

- Portfolio Weights
 - The fraction of the total investment in the portfolio held in each individual investment in the portfolio
 - The portfolio weights must add up to 1.00 or 100%.

$$x_i = \frac{\text{Value of investment } i}{\text{Total value of portfolio}}$$

- Portfolio Return

$$R_P = x_1R_1 + x_2R_2 + \cdots + x_nR_n = \sum_i x_iR_i$$

- Expected return on a portfolio

$$E[R_P] = E\left[\sum_i x_iR_i\right] = \sum_i E[x_iR_i] = \sum_i x_iE[R_i]$$

Covariance and Correlation

- Covariance
 - The expected product of the deviations of two returns from their means
 - Covariance between Returns R_i and R_j

$$\text{Cov}(R_i, R_j) = E[(R_i - E[R_i])(R_j - E[R_j])]$$

- Estimate of the Covariance from Historical Data

$$\text{Cov}(R_i, R_j) = \frac{1}{T-1} \sum_t (R_{i,t} - \bar{R}_i)(R_{j,t} - \bar{R}_j)$$

- If the covariance is positive, the two returns tend to move together.
- If the covariance is negative, the two returns tend to move in opposite directions.

Covariance and Correlation (2 of 2)

- Correlation
 - A measure of the common risk shared by stocks that does not depend on their volatility

$$\text{Corr}(R_i, R_j) = \frac{\text{Cov}(R_i, R_j)}{\text{SD}(R_i) \text{SD}(R_j)}$$

- The correlation between two stocks will always be between -1 and $+1$.
- The Covariance and Correlation of a Stock with Itself

$$\text{Cov}(R_s, R_s) = E[(R_s - E[R_s])(R_s - E[R_s])] = E[(R_s - E[R_s])^2] = \text{Var}(R_s)$$

$$\text{Corr}(R_s, R_s) = \frac{\text{Cov}(R_s, R_s)}{\text{SD}(R_s) \text{SD}(R_s)} = \frac{\text{Var}(R_s)}{\text{SD}(R_s)^2} = 1$$

Portfolio's Variance and Volatility

- For a two security portfolio,

$$\begin{aligned} \text{Var}(R_P) &= \text{Cov}(R_P, R_P) \\ &= \text{Cov}(x_1 R_1 + x_2 R_2, x_1 R_1 + x_2 R_2) \\ &= x_1 x_1 \text{Cov}(R_1, R_1) + x_1 x_2 \text{Cov}(R_1, R_2) + x_2 x_1 \text{Cov}(R_2, R_1) + x_2 x_2 \text{Cov}(R_2, R_2) \end{aligned}$$

- The Variance of a Two-Stock Portfolio

$$\text{Var}(R_P) = x_1^2 \text{Var}(R_1) + x_2^2 \text{Var}(R_2) + 2x_1 x_2 \text{Cov}(R_1, R_2)$$

- For a large portfolio

$$\text{Var}(R_P) = \text{Cov}(R_P, R_P) = \text{Cov}\left(\sum_i x_i R_i, R_P\right) = \sum_i x_i \text{Cov}(R_i, R_P)$$

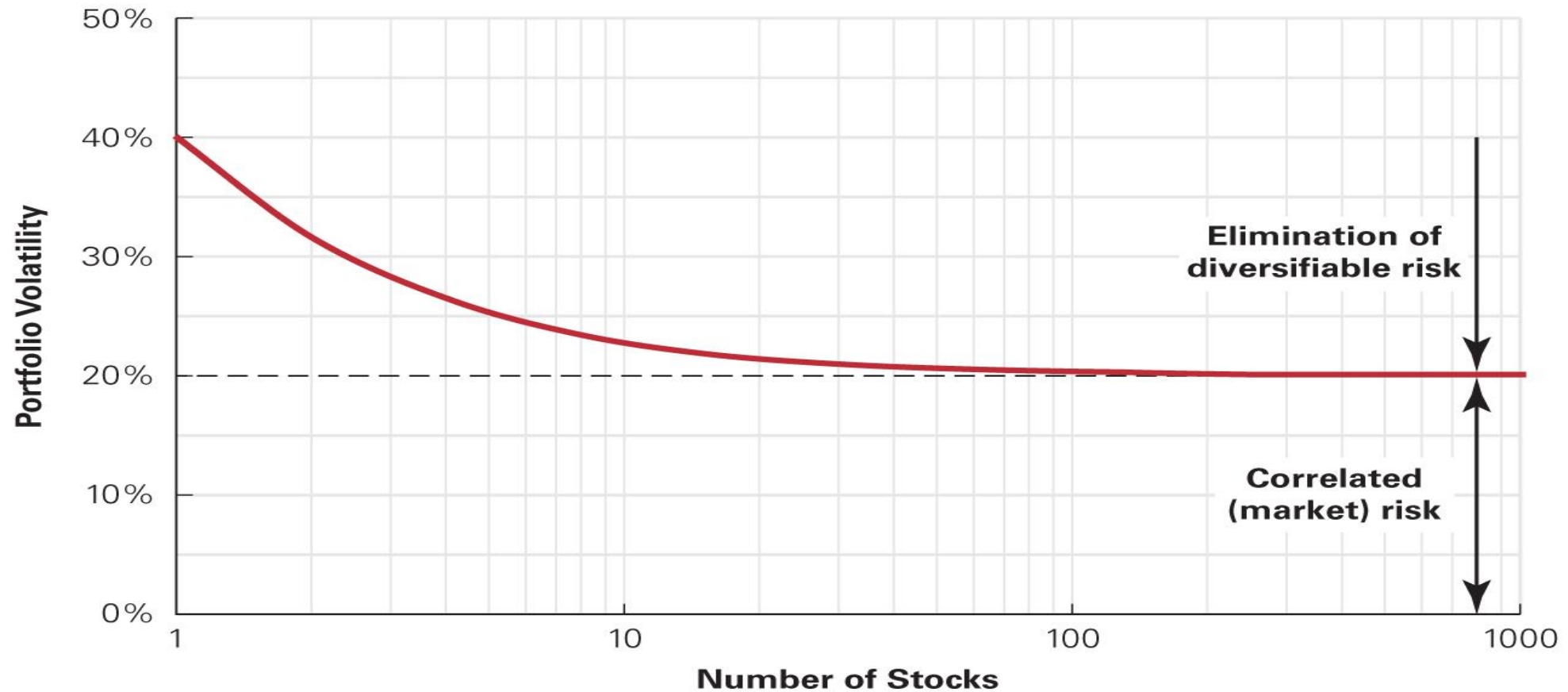
$$\begin{aligned} \text{Var}(R_P) &= \sum_i x_i \text{Cov}(R_i, R_P) = \sum_i x_i \text{Cov}(R_i, \sum_j x_j R_j) \\ &= \sum_i \sum_j x_i x_j \text{Cov}(R_i, R_j) \end{aligned}$$

Equally Weighted Portfolio

- Equally Weighted Portfolio
 - A portfolio in which the same amount is invested in each stock
- Variance of an Equally Weighted Portfolio of n Stocks

$$\begin{aligned} \text{Var}(R_p) = & \frac{1}{n} (\text{Average Variance of the Individual Stocks}) \\ & + \left(1 - \frac{1}{n}\right) (\text{Average Covariance between the Stocks}) \end{aligned}$$

Volatility of an Equally Weighted Portfolio Versus the Number of Stocks



Diversification with General Portfolios

- For a portfolio with arbitrary weights, the standard deviation is calculated as follows:
 - Volatility of a Portfolio with Arbitrary Weights

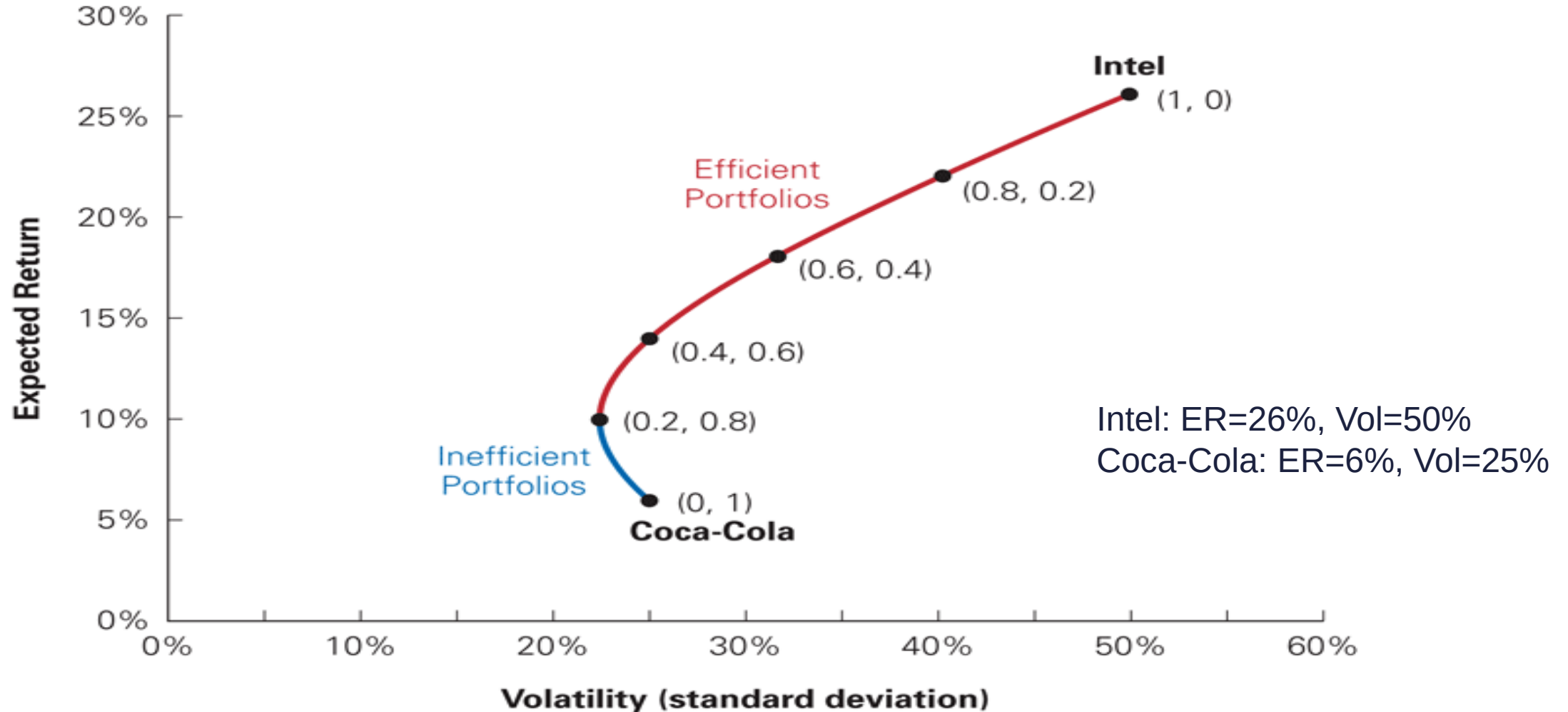
$$SD(R_P) = \sum_i \overbrace{x_i \times SD(R_i) \times Corr(R_i, R_P)}^{\text{Security } i\text{'s contribution to the volatility of the portfolio}}$$

$\underbrace{\hspace{1cm}}_{\text{Amount of } i \text{ held}} \quad \underbrace{\hspace{1cm}}_{\text{Total risk of } i} \quad \underbrace{\hspace{1cm}}_{\text{Fraction of } i\text{'s risk that is common to } P}$

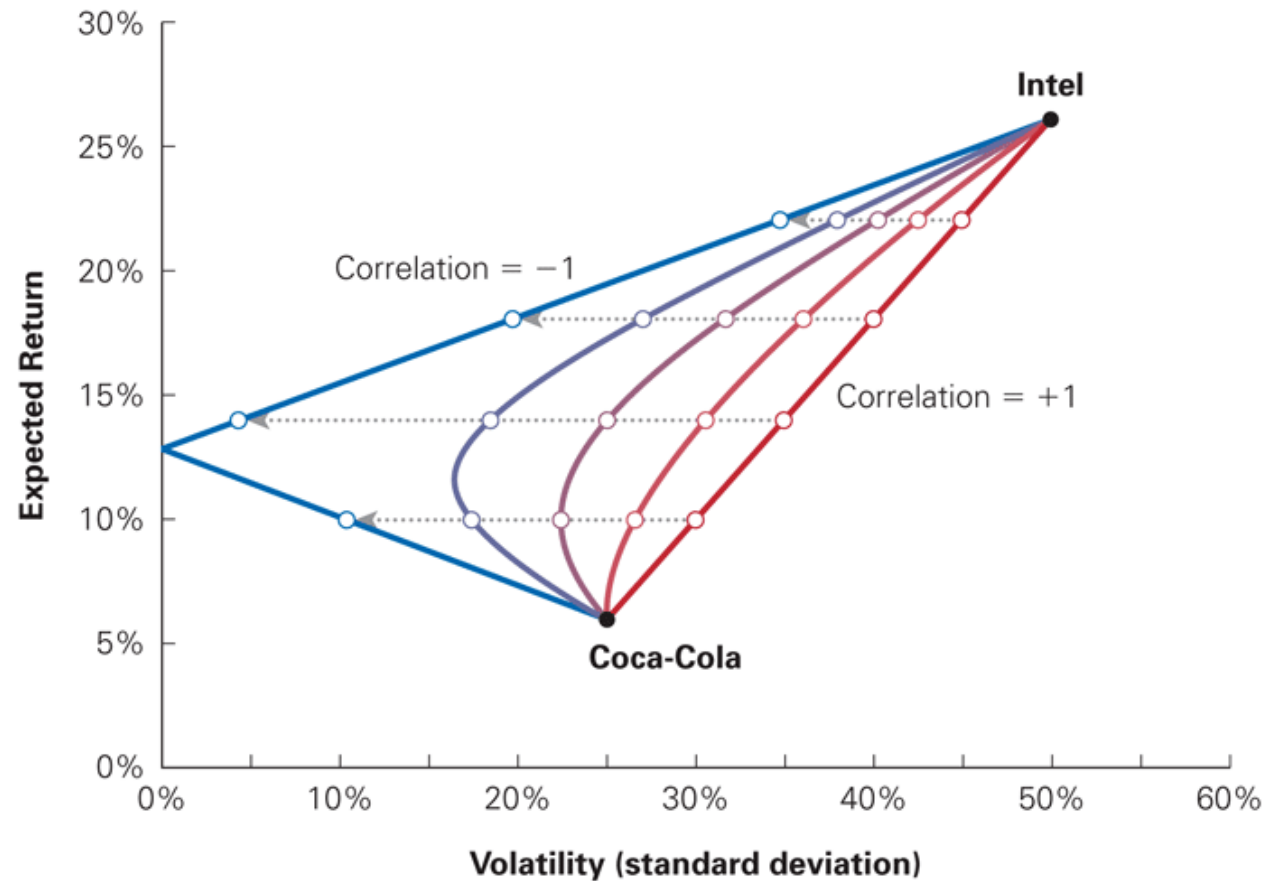
- Unless all of the stocks in a portfolio have a perfect positive correlation of +1 with one another, the risk of the portfolio will be lower than the weighted average volatility of the individual stocks:

$$SD(R_P) = \sum_i x_i SD(R_i) Corr(R_i, R_P) < \sum_i x_i SD(R_i)$$

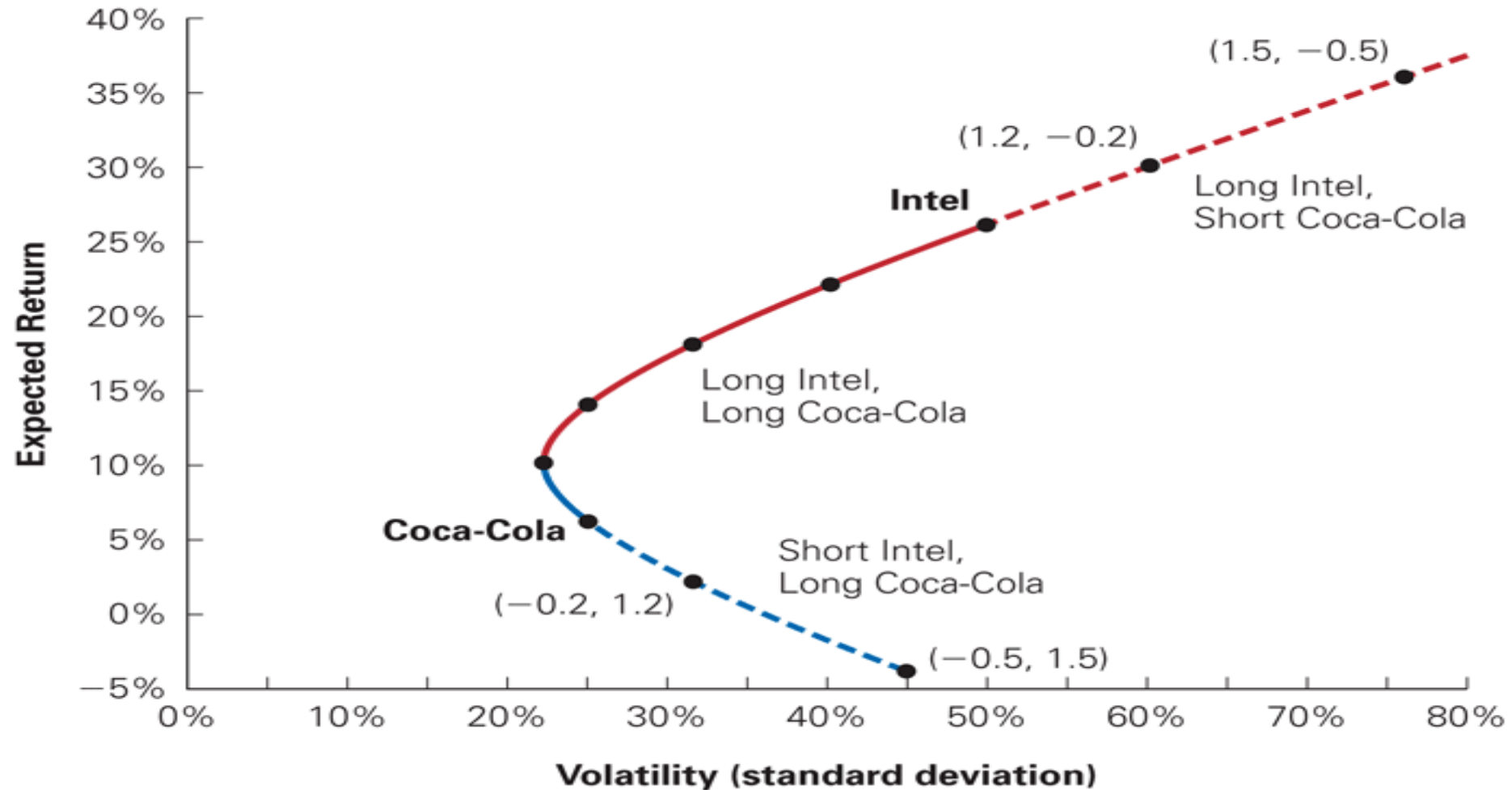
Volatility Versus Expected Return for Portfolios of Intel and Coca-Cola Stock



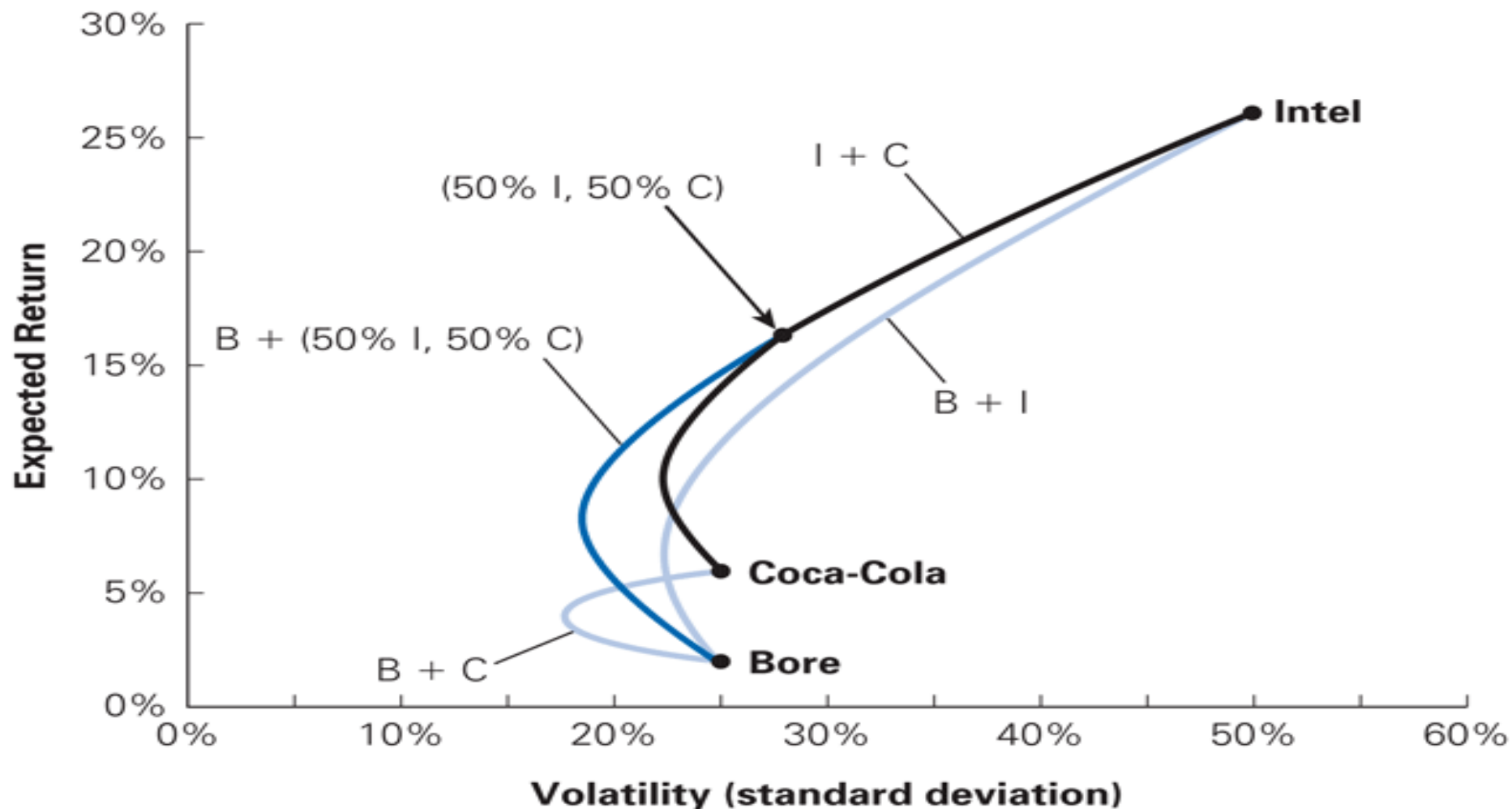
The Effect of correlation



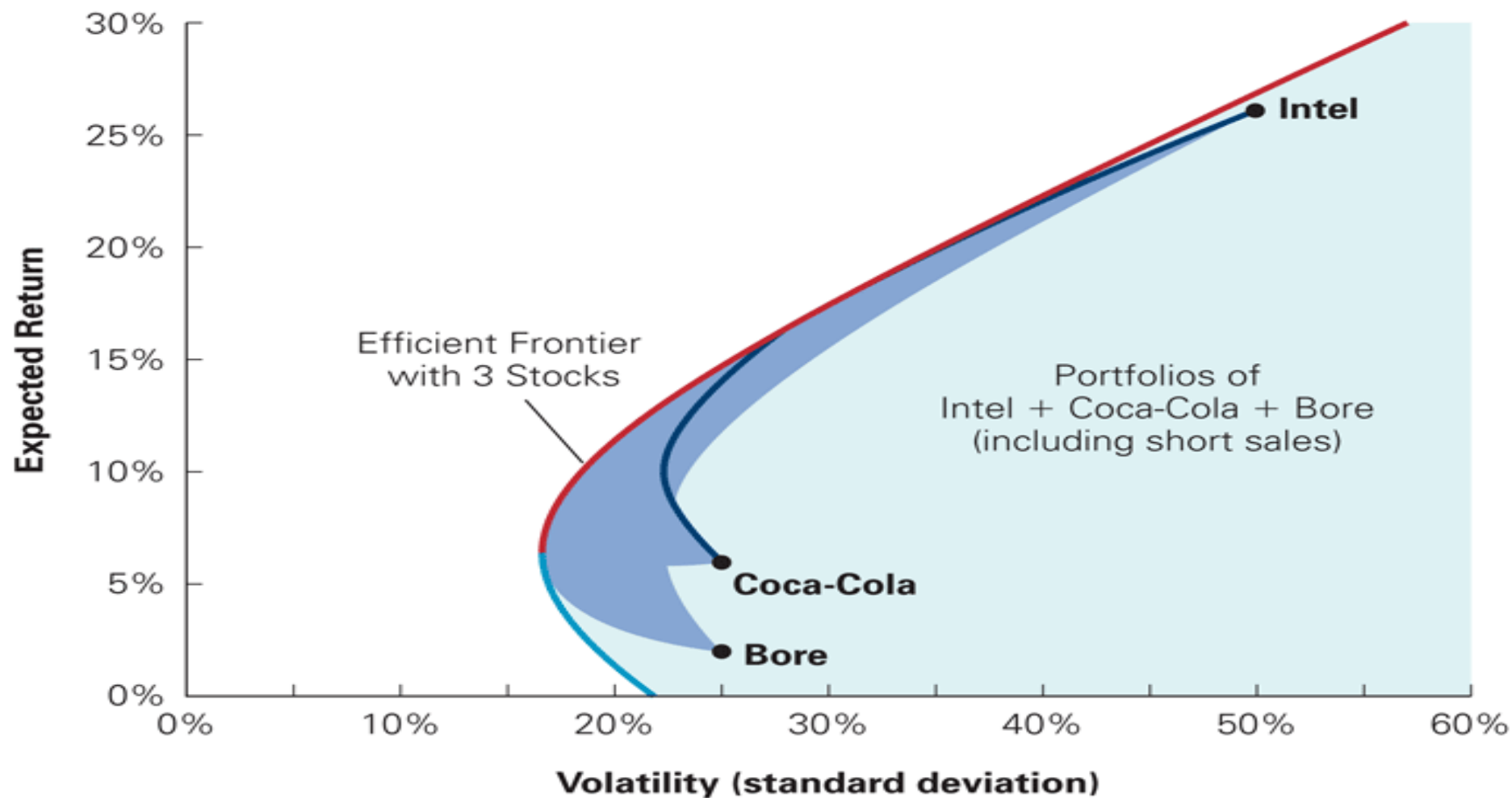
Allowing for Short Sales



Expected Return and Volatility for Selected Portfolios of Intel, Coca-Cola, and Bore Industries Stocks



The Volatility and Expected Return for All Portfolios of Intel, Coca-Cola, and Bore Stock



- A fraction of the money is in a risky portfolio, while the remaining fraction in the risk-free Treasury bills.
 - The expected return would be

$$\begin{aligned} E[R_{xP}] &= (1-x)r_f + xE[R_P] \\ &= r_f + x(E[R_P] - r_f) \end{aligned}$$

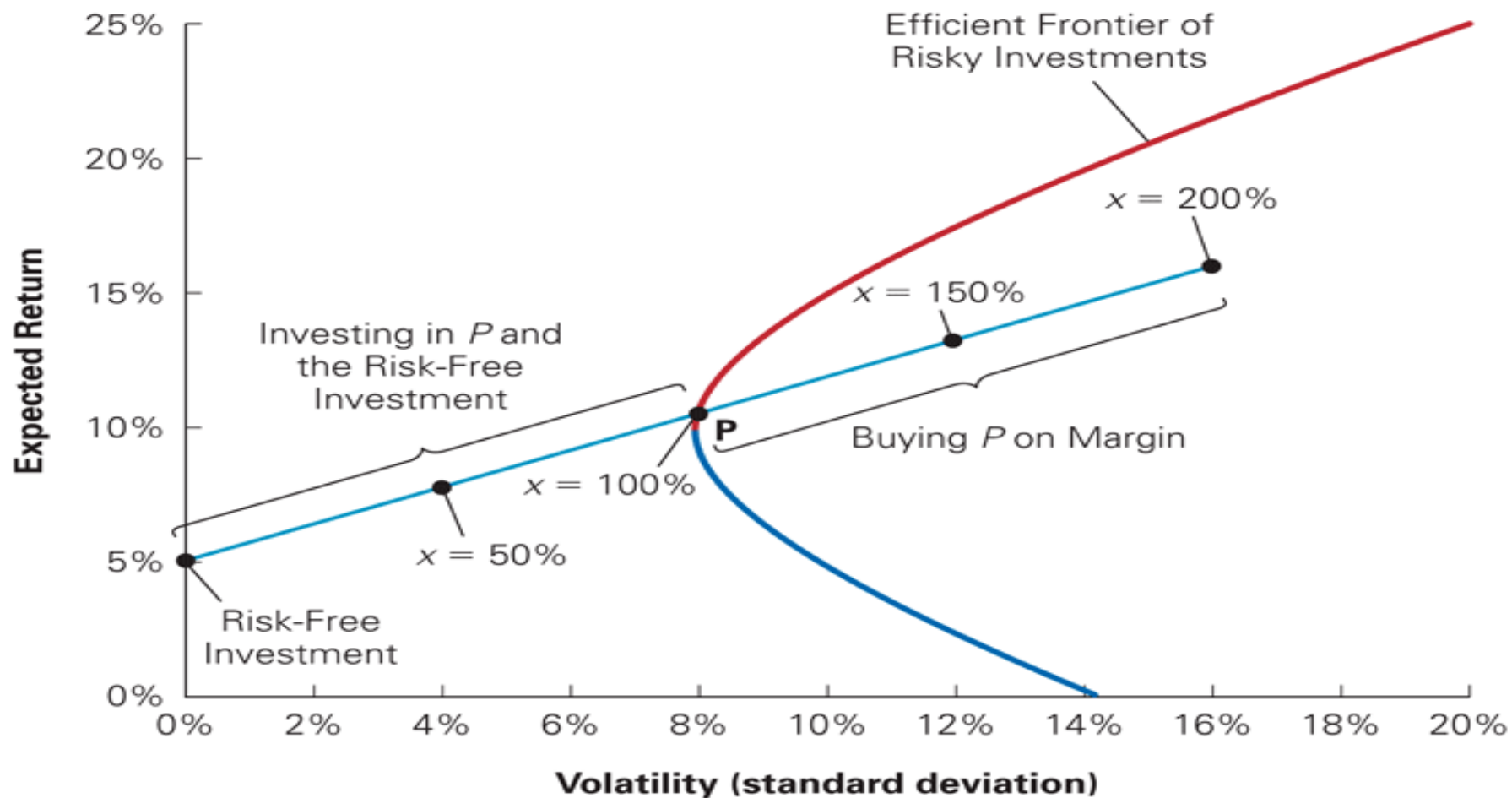
- The standard deviation of the portfolio would be calculated as follows:

$$\begin{aligned} SD[R_{xP}] &= \sqrt{(1-x)^2 \text{Var}(r_f) + x^2 \text{Var}(R_P) + 2(1-x)x \text{Cov}(r_f, R_P)} \\ &= \sqrt{x^2 \text{Var}(R_P)} \\ &= xSD(R_P) \end{aligned}$$

0



Combining a Risk-Free Investment and a Risky Portfolio



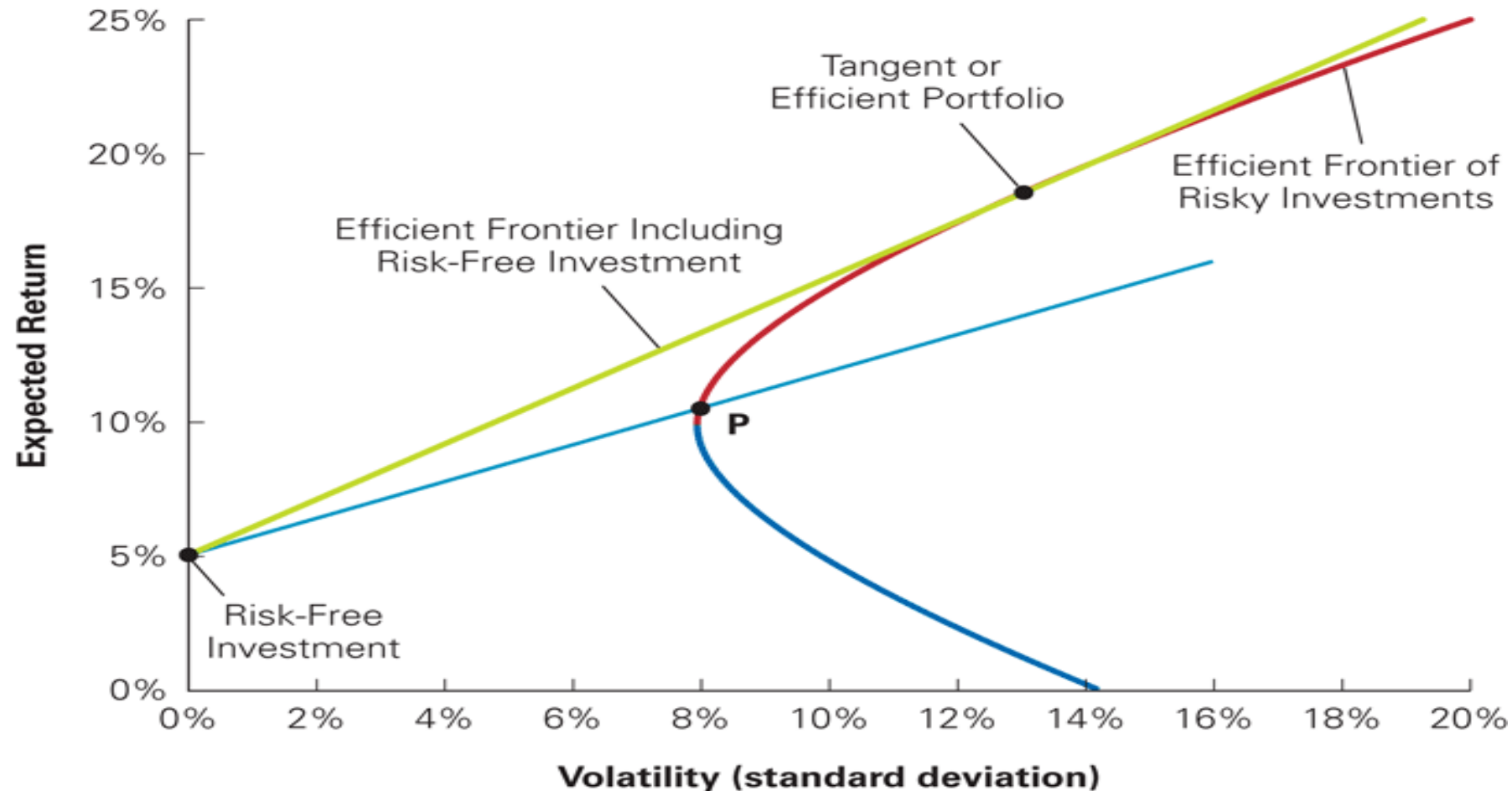
Identifying the Tangent Portfolio

- Sharpe Ratio
 - Measures the ratio of reward-to-volatility provided by a portfolio

$$\text{Sharpe Ratio} = \frac{\text{Portfolio Excess Return}}{\text{Portfolio Volatility}} = \frac{E[R_P] - r_f}{SD(R_P)}$$

- The portfolio with the highest Sharpe ratio is the portfolio where the line with the risk-free investment is tangent to the efficient frontier of risky investments. The portfolio that generates this tangent line is known as the **tangent portfolio**.

The Tangent or Efficient Portfolio



- Portfolio Improvement: Beta and the Required Return

$$\underbrace{E[R_i] - r_f}_{\text{Additional return from investment i}} > \underbrace{SD(R_i) \times \text{Corr}(R_i, R_P)}_{\text{Incremental volatility from investment i}} \times \underbrace{\frac{E[R_P] - r_f}{SD(R_P)}}_{\text{Return per unit of volatility available from portfolio P}}$$

- Beta of Portfolio i with Portfolio P

$$\beta_i^P = \frac{SD(R_i) \times \text{Corr}(R_i, R_P)}{SD(R_P)} = \frac{\text{Cov}(R_i, R_P)}{\text{Var}(R_P)}$$

- the Required Return

$$r_i = r_f + \beta_i^P \times (E[R_P] - r_f)$$

Efficient portfolio and CAPM^{9/20/2024}

- Expected Return of a Security

$$E[R_i] = r_f + \beta_i^{eff} \times (E[R_{eff}] - r_f)$$

- The Capital Asset Pricing Model (CAPM) allows us to identify the efficient portfolio of risky assets without having any knowledge of the expected return of each security.
- Instead, the CAPM uses the optimal choices investors make to identify the efficient portfolio as the market portfolio, the portfolio of all stocks and securities in the market.

The CAPM Assumptions

1. Investors can buy and sell all securities at competitive market prices (without incurring taxes or transactions costs) and can borrow and lend at the risk-free interest rate.
2. Investors hold only efficient portfolios of traded securities—portfolios that yield the maximum expected return for a given level of volatility.
3. Investors have **homogeneous expectations** regarding the volatilities, correlations, and expected returns of securities.
 - Homogeneous Expectations
 - All investors have the same estimates concerning future investments and returns.

Supply, Demand, and the Efficiency of the Market Portfolio

- Given homogeneous expectations, all investors will demand the same efficient portfolio of risky securities.
- The combined portfolio of risky securities of all investors must equal the efficient portfolio.
- Thus, if all investors demand the efficient portfolio, and the supply of securities is the market portfolio, the demand for market portfolio must equal the supply of the market portfolio.

Supply, Demand, and the Efficiency of the Market Portfolio

- Given homogeneous expectations, all investors will demand the same efficient portfolio of risky securities.
- The combined portfolio of risky securities of all investors must equal the efficient portfolio.
- Thus, if all investors demand the efficient portfolio, and the supply of securities is the market portfolio, the demand for market portfolio must equal the supply of the market portfolio.
- When the CAPM assumptions hold, an optimal portfolio is a combination of the risk-free investment and the market portfolio.
 - When the tangent line goes through the market portfolio, it is called the **capital market line (CML)**.

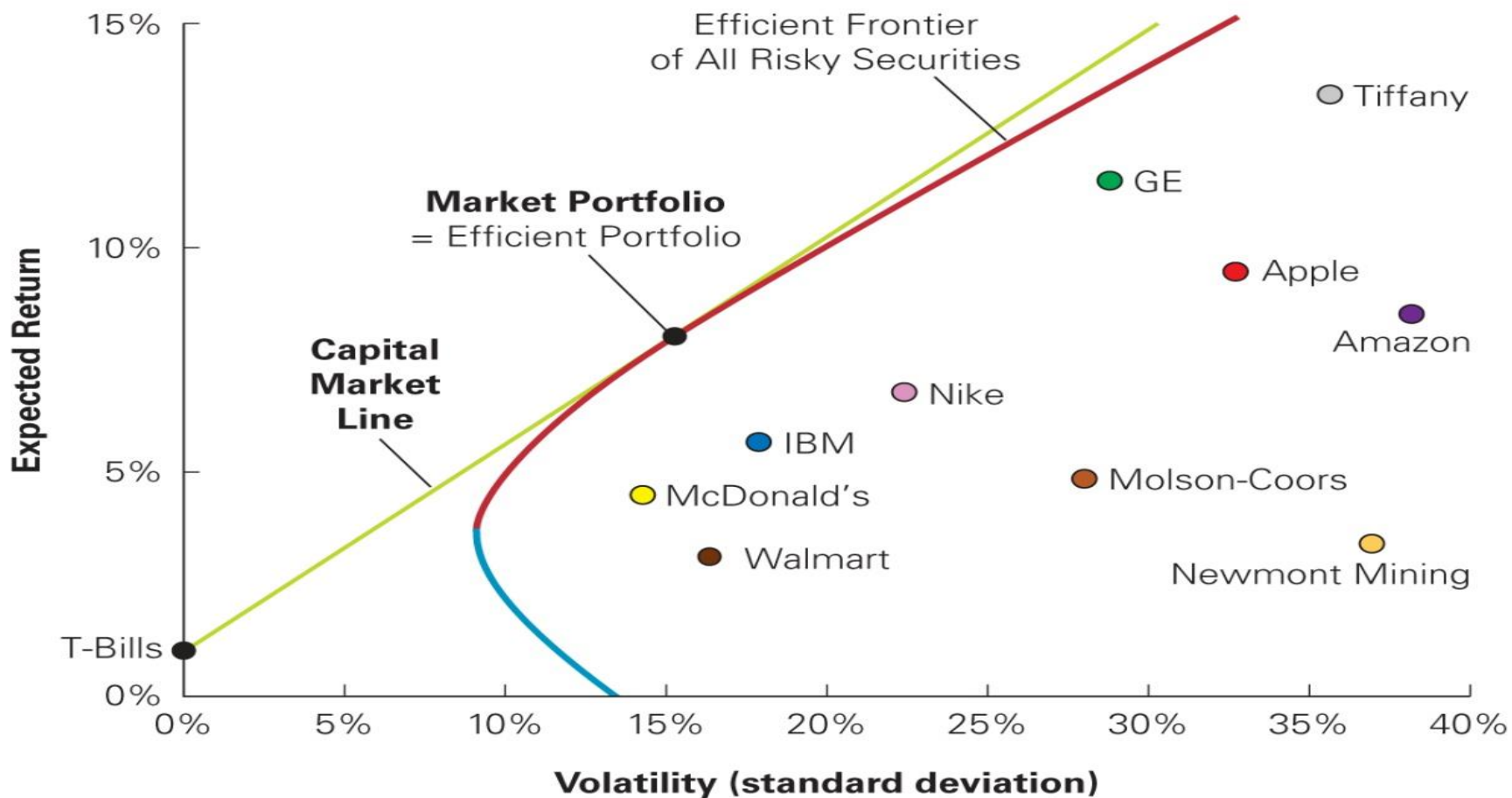
Optimal Investing: The Capital Market Line

- The expected return and volatility of a capital market line portfolio are

$$E[R_{xCML}] = (1 - x)r_f + xE[R_{Mkt}] = r_f + x(E[R_{Mkt}] - r_f)$$

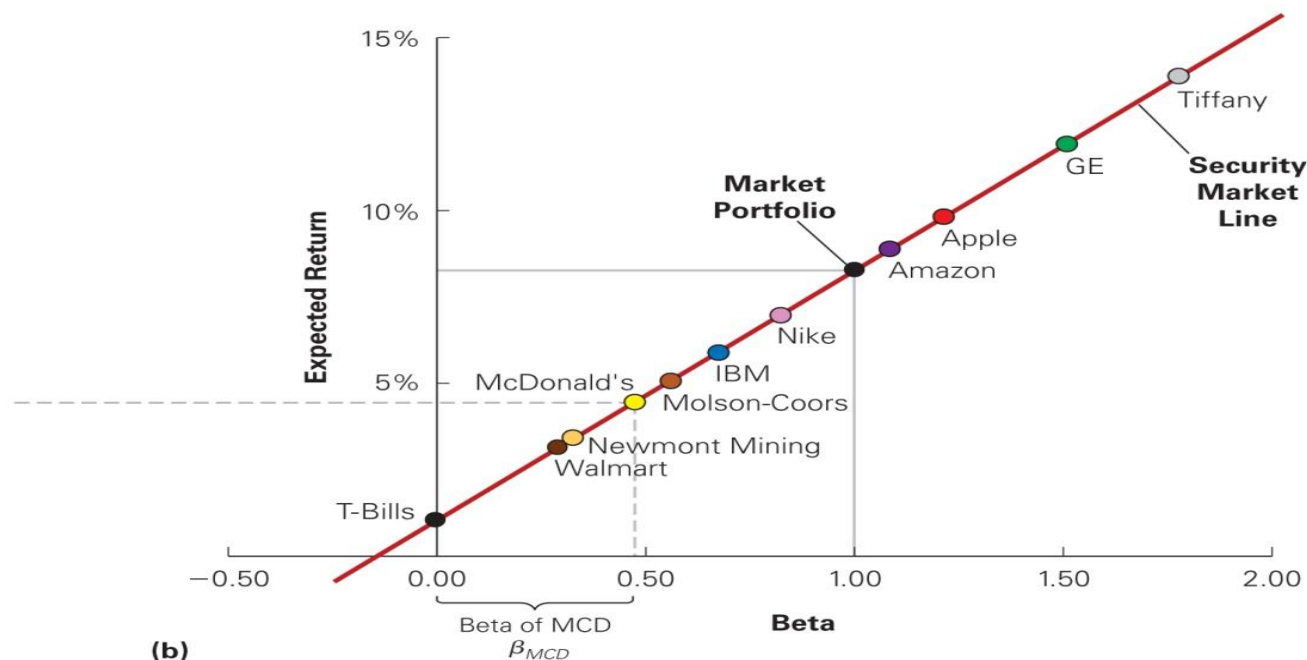
$$SD(R_{xCML}) = xSD(R_{Mkt})$$

The Capital Market Line



The Capital Market Line and the Security Market Line

The SML shows the expected return for each security as a function of its beta with the market. According to the CAPM, the market portfolio is efficient, so all stocks and portfolios should lie on the SML.



The Security Market Line

- Beta of a Portfolio
 - The beta of a portfolio is the weighted average beta of the securities in the portfolio.

$$\beta_P = \frac{Cov(R_P, R_{Mkt})}{Var(R_{Mkt})} = \frac{Cov\left(\sum_i x_i R_i, R_{Mkt}\right)}{Var(R_{Mkt})} = \sum_i x_i \frac{Cov(R_i, R_{Mkt})}{Var(R_{Mkt})} = \sum_i x_i \beta_i$$

Patterns of Corporate Financing

Firms may raise funds through
external sources or
plow back profits rather than distribute them to shareholders.

Should a firm elect external financing, they may choose between
debt or
equity sources.

Equity - Common Stock

Book Value vs. Market Value

- Book value is a backward looking measure. It tells us how much capital the firm has raised from shareholders in the past. It does not measure the value that shareholders place on those shares today.
- Market value of the equity is forward looking, it depends on the future dividends that shareholders expect to receive.

Debt Instruments

Whom the loan is borrowed from:

- Bank

- Issue bonds

Maturity:

- Short term (usually it is rolled over)

- Long term

Collaterals → lending against assets or against cash-flow (unsecured debt)

Covenants: Company can transfer control rights to lenders through covenants

Priority among debt instruments → senior debt vs. subordinated debt

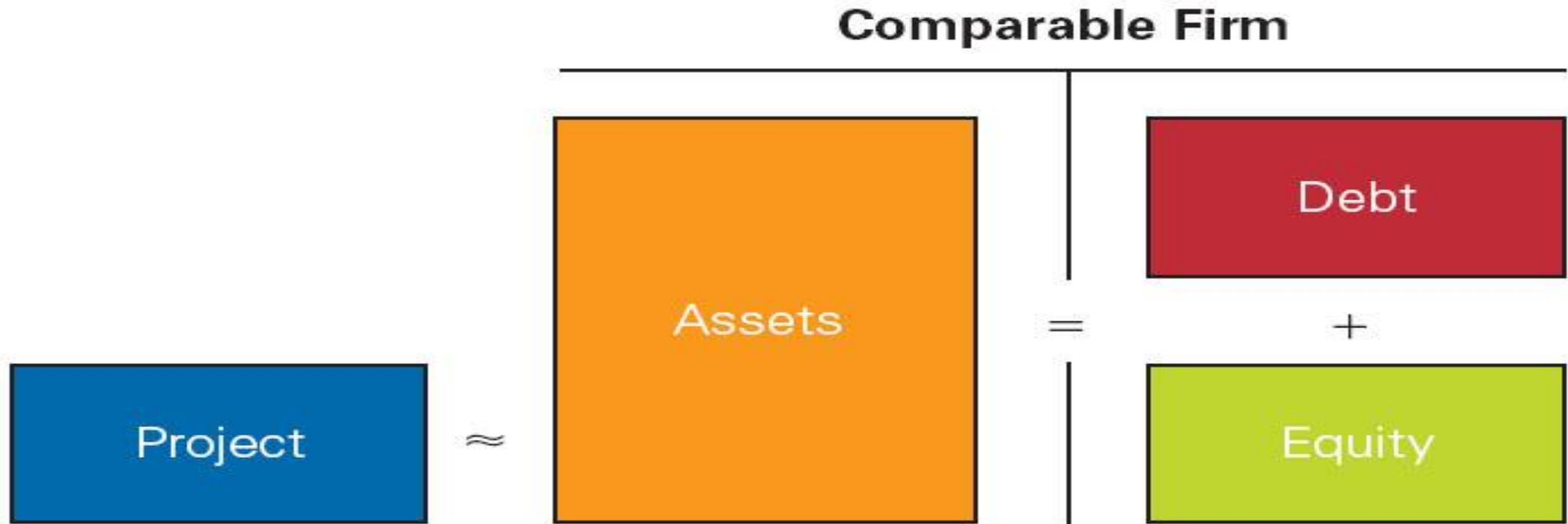
Other Types of Securities

Warrant - Right to buy shares from a company at a stipulated price before a set date.

Convertible Bond - Bond that the holder may exchange for a specified amount of another security.

Callable Bond - Bond that may be repurchased by firm before maturity at specified call price.

Using a Levered Firm as a Comparable for a Project's Risk



A Project's Cost of Capital

- Asset (unlevered) cost of capital
 - Expected return required by investors to hold the firm's underlying assets.
 - Weighted average of the firm's equity and debt costs of capital

$$r_U = \frac{E}{E + D} r_E + \frac{D}{E + D} r_D$$

- Asset (unlevered) beta

$$\beta_U = \frac{E}{E + D} \beta_E + \frac{D}{E + D} \beta_D$$

The debt cost of capital

- Debt Yields Versus Returns
 - Yield to maturity is the IRR an investor will earn from holding the bond to maturity and receiving its promised payments.
 - If there is little risk the firm will default, yield to maturity is a reasonable estimate of investors' expected rate of return.
 - If there is significant risk of default, yield to maturity will overstate investors' expected return.

The debt cost of capital

- Consider a one-year bond with YTM of y . For each \$1 invested in the bond today, the issuer promises to pay $\$(1 + y)$ in one year.
- Suppose the bond will default with probability p , in which case bond holders receive only $\$(1 + y - L)$, where L is the expected loss per \$1 of debt in the event of default.

- So the expected return of the bond is

$$r_d = (1 - p)y + p(y - L) = y - pL$$

= Yield to Maturity – Prob(default) × Expected Loss Rate

- The importance of the adjustment depends on the riskiness of the bond.

The debt cost of capital

- Annual Default Rates by Debt Rating (1983–2011)

Rating:	AAA	AA	A	BBB	BB	B	CCC	CC–C
Default Rate:								
Average	0.0%	0.1%	0.2%	0.5%	2.2%	5.5%	12.2%	14.1%
In Recessions	0.0%	1.0%	3.0%	3.0%	8.0%	16.0%	48.0%	79.0%

Source: “Corporate Defaults and Recovery Rates, 1920-2011,” **Moody’s Global Credit Policy**, February 2012